



Choroidal melanoma

Dose–response of critical structures in the posterior eye segment to hypofractionated stereotactic photon radiotherapy of choroidal melanoma



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ABSTRACT

Purpose: To identify modifying factors and dose–/volume–response relationships for the retina and optic nerve related to highly conformal hypofractionated radiotherapy.

Patients and methods: Seventy-three patients undergoing hypofractionated stereotactic photon radiotherapy of choroidal melanoma were included in this retrospective study.

The volumes of the optic nerve receiving doses of more than 7.5 or 12 Gy, respectively, were defined. Optic nerve circumference included in the 30%, 40%, 50%, or 80% isodose (ON_x) and retina included in 30% or 40% were determined as quantal effects.

Univariate and multivariate analyses were performed for clinical variables as well as probit analysis to define ED_x (doses where a positive response is expected in *x*% of the cases).

Results: Median follow-up was 90.0 (interquartile range 69.0–98.0) months. Fifty-two (71%) and 49 (67%) patients developed radiation retinopathy and optic neuropathy (any grade). Age, length of follow-up and diabetes were significant parameters regarding retinopathy. Optic neuropathy was significantly influenced by age, length of follow-up, and ON₃₀.

The probability of optic neuropathy (any grade and grade ≥ 2) significantly increases with the dose (*p* ranges from 0.0126 to 0.0211).

Conclusion: Treatment planning should aim at minimizing encompassing isodoses particularly in the low dose region, without compromising PTV coverage.

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Choroidal (uveal) melanoma is the most common intraocular malignancy in adults [7]. Besides surgical intervention, radiotherapy is a highly effective curative primary treatment modality. As an alternative to brachytherapy or external beam proton therapy, stereotactic radiotherapy with high-energy photons may be administered [4–6,12,14]. At the Medical University of Vienna linear accelerator-based hypofractionated stereotactic radiotherapy is applied in patients with posterior tumor localization unsuitable for brachytherapy or surgical resection [5,6]. High precision target positioning is guaranteed with a dedicated automated eye monitoring and surveillance system. This system was designed to be utilized already during imaging for treatment planning (CT- and MR) as well as during treatment delivery [2,16]. With this image-guided stereotactic radiotherapy technique using online target position monitoring, high local tumor control rates of 95.9% after

5 years and 92.6% after 10 years, respectively, were achieved, with acceptable treatment-related morbidity rates [5,6].

As this treatment technique was based on three-dimensional imaging and three-dimensional dose calculation, dose–volume information can be extracted retrospectively from treatment plans [8]. Furthermore, ocular structures involved in the treatment of uveal melanoma are rather stable in terms of geometry. Spatially invariant structures and dose distributions facilitated the correlation of observed side effects with dose and dose–volume parameters. As far as dose–volume effects of the optic nerve and other posterior parts of the eye are concerned, literature is limited. The QUANTEC chapter on the optic nerve only deals with standard fractionation schemes or single fraction therapy [11], concluding in existing literature dose reporting for the optic nerve is generally limited to the maximum dose. There is a lack of data from hypofractionated treatments for the optic nerve and other ocular structures in general.

The aim of the present study was therefore to identify modifying factors and to establish dose–/volume–response relationships for critical structures in the posterior segment of the eye (retina

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